TPC-415: Three-Shell Height Extension

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Abstract

We extend the three-shell pooled complete-shell proxy to $H = 16, 32, 66, 128$, with $N = 4H$. The shells at $Q = 65536, 131072, 262144$ retain 36848 primes, use shell-local amplitudes, and have equal pooled parity counts 18424/18424. Exact arithmetic and an independent literal replay verify all four rows. This is finite synthetic evidence, not an arithmetic theorem.

1 Profile and bound

Let $\mathcal{P}$ be the increasing union of $(65536, 131072]$, $(131072, 262144]$, and $(262144, 524288]$. For $p_i \in \mathcal{P}$, set $a_i = p_i^3/[Q_i^2(p_i - 1)]$ using its source-shell scale. For each listed H, put $N = 4H$ and use alternating CRT residues. All primes exceed the largest $N = 512$, so

$$G_0 = V_- S_0, \qquad G_1 = V_- S_1 + V_+(S_1 - t_1^2), \qquad M = t_1 P_-.$$

The pooled count is 36848, hence $m_- = m_+ = 18424$. Cauchy–Schwarz and $S_0, S_1 \geq H/4$ give

$$0 \leq z \leq \frac{t_1}{a_{\min}\sqrt{S_0 S_1}} \leq \frac{4}{a_{\min} H} \leq \frac{4}{H}. \tag{1}$$

2 Exact observations

The certificate contains all prime and source-shell labels and exact rational rows. The normalized observations at the four heights are, to seven decimal places, 0.0217692, 0.0108013, 0.0052131, and 0.0026820. The independent checker rebuilds the three sieves and literally visits every mask for every prime and coordinate before comparing G_0, G_1, and M.

3 Scope and reproduction

This finite four-height synthetic adjacent proxy does not prove a full operator estimate, physical h_0 or arithmetic signs, pay arithmetic L^2 or fixed-power credit, close Route A/B, or imply a twin-prime result. Run the producer, replay, and stress with `--check` in normal and optimized modes; Bridge-B repeats and locks these checks.